

Comprehensive Analysis of Rice Crop Identification and Consumer Digital Twin Development

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bjahagir-abhingle-shahnam-nsantha.git](https://github.iu.edu/Luddy-B565-SP24/bjahagir-abhingle-shahnam-nsantha.git)

Abstract

In this report, we discuss two projects we've been working on. Firstly, we developed a model to identify where rice is grown in Vietnam using satellite images data. This helps us better plan for food production and ensure food security. Secondly, we created another model to analyze how people shop for groceries. Using this model, we aim to provide personalized recommendations to shoppers. We tested various methods and achieved results. These projects highlight the potential of technology, such as machine learning and data analysis, to address challenges in agriculture and retail.

We believe that leveraging technology in these sectors can lead to significant improvements and innovations, benefiting both producers and consumers.

1 Introduction

The demand for efficient agricultural practices has grown significantly in recent years, driven by the need to address global food security challenges. In this context, the accurate identification of crop types, such as rice, plays a crucial role in optimizing agricultural strategies and resource allocation. This report focuses on the development and evaluation of a rice crop identification model for 250 geo-locations in the An Giang province of Vietnam using satellite data. The significance of this study lies in its potential to contribute to the prioritization of actions aimed at tackling hunger by identifying areas where rice, a staple food crop for many populations, is growing. By leveraging geospatial datasets and machine learning techniques, our research aims to create a classification model capable of distinguishing between rice and non-rice fields with high precision. The approach taken in this study involves enhancing model accuracy by leveraging feature values with higher correlation and extracting predictor variable values for the entire year, rather than relying on single-date data. By comparing our model with existing solutions and refining parameters such as C, penalty, solver, and class weight, we seek to demonstrate improvements in crop identification performance.

In parallel, our team embarked on a secondary project focused on the consumer sector. Here, the development of a Consumer Digital Twin using grocery shopping data aimed to build a digital representation of consumers based on their shopping patterns. This initiative looks to improve retail strategies and personalize marketing efforts, leveraging data analysis to predict consumer behavior and recommend products, thereby enhancing customer experiences and business profitability. This introduction sets the stage for a dual-focused report that discusses both agricultural innovations and advances in consumer analytics.

2 Related Work

The field of crop identification and classification using remote sensing data has been the subject of extensive research. Several studies have explored methodologies and approaches that align closely with our proposed solution, offering valuable insights and comparisons. Lobell, D.B. delves into methodologies for crop identification using satellite imagery, highlighting the importance of feature extraction and machine learning algorithms in achieving accurate classification results. Their work provides a foundation for our approach, particularly in terms of leveraging

feature values with higher correlation to enhance model precision. Hao, P., Wang, L. contributes crucial insights into potential pitfalls and challenges encountered in similar studies. By addressing issues such as data variability, model overfitting, and feature selection, Jones’ work serves as a reference point for refining our methodology and ensuring robustness in our approach. Additionally, studies by researchers have explored the use of different machine learning algorithms, including support vector machines (SVMs) and random forests, for crop classification tasks. These studies provide valuable comparisons and benchmarks for evaluating the performance of our model against alternative approaches.

Parallel to our agricultural research, the exploration of consumer digital twins using grocery shopping data is informed by an emerging body of work in the fields of data science and personalized marketing. Garanayak, Mamata, Mohanty, Sachi, Jagadev, Alok, and Sahoo, Sipra delve into methodologies that closely align with our proposed solution for recommendation systems based on collaborative filtering and item-based similarity. Their work emphasizes the importance of cosine similarity and the KNN Basic model in generating accurate item recommendations, which resonates with our approach. Phorasim and L. Yu provide crucial insights into potential pitfalls and challenges faced in similar studies related to recommendation systems and data preprocessing. Their examination of data biases, model evaluation techniques, and algorithmic performance is instrumental in guiding our methodology and addressing key issues in our project. S. Gong presents a comprehensive review of item-based collaborative filtering algorithms and their applications in recommendation systems. Their study highlights the significance of feature engineering, similarity metrics, and model evaluation in optimizing recommendation accuracy, which aligns with our focus on item-item similarities and model effectiveness.

By reviewing and synthesizing existing literature, we aim to build upon previous research findings and contribute novel insights to the field of crop identification using geospatial datasets and machine learning techniques as well as the development of digital consumer profiles through advanced data analysis techniques.

3 Methodology

3.1 Rice Crop Identification

The methodology employed in the rice crop identification study revolves around developing a classification model using satellite data from the Sentinel-1 dataset. Key steps performed are outlined below.

3.1.1 Data Collection

Given a Dataset with geo-location data (Latitude and Longitude) with tags specifying the presence of rice or non-rice crops from a specific region in Vietnam for the year 2020. Extracted variables such as VV and VH for an entire year, calculated their mean from the Sentinel-1 RTC dataset and computed RVI AND NDVI values using extracted variables and used them as predictor variables as they have higher correlation values. While extracting the predictor variables used 3x3, 5x5, or 7x7 window around the specific latitude and longitude point to get improved results.

3.1.2 Data Preprocessing

We implemented feature scaling using the Standard Scaler to normalize the VV and VH variables, which is crucial for many machine learning algorithms. The data was split into 70

3.1.3 Model Selection and Training

We opted for a binary logistic regression model from the scikit-learn library due to the predictor variable value being categorical (Rice or Non-Rice). The model was trained using the training data, with predictor variables (RVI and NDVI) and response variable (presence of rice crop) separated into appropriate arrays (X and Y).

3.1.4 Model Evaluation

In-sample evaluation involved generating a classification report and confusion matrix using the training data to assess the model's performance. Out-sample evaluation was conducted by predicting on the test data to estimate the model's ability to generalize and avoid overfitting.

3.2 Consumer Digital Twin Development

The methodology for creating a Consumer Digital Twin using grocery shopping data encompassed several stages, including data collection, preprocessing, model development, and validation.

3.2.1 Data Collection

For the purpose of this experiment, we could not find any datasets that are publicly available. Hence, we curated a dataset to replicate human shopping behavior.

We generated a dataset for a single consumer. To replicate human behavior, we designed our code in such a way that the user is shopping from three categories, namely: essentials, frequently bought, and rarely bought items. The essential items, which include items like milk, eggs, bread, etc., are bought every 3-5 days. Frequently bought items are purchased every 10-15 days, and rarely bought items are purchased every 1-3 months.

3.2.2 Data Preprocessing

From the curated dataset, we did not have any outliers or missing values. We performed feature engineering and created two additional features for each entry, namely: Previous Purchase Date and Days After Previous Purchase. We handled categorical variables and date columns as required by the models.

3.2.3 Model Development, Model Training and Validation

We developed 3 models to create a Consumer Digital Twin:

- KNN
- Random Forest
- GANs

3.2.4 KNN

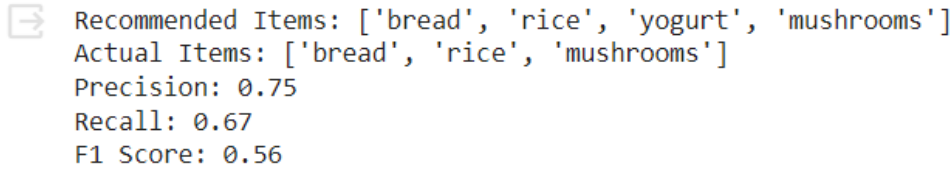
3.2.5 Similarity Calculation

To implement item-based collaborative filtering, we calculated item-item similarities using cosine similarity. This involved Transforming the dataset into a suitable format for similarity calculations. Using the cosine similarity metric to measure similarities between items based on their attributes.

3.2.6 KNN Basic Model

We employed the KNN Basic model for item-based collaborative filtering, because it utilizes item-item similarities for recommendations, is simple to implement and understand, and can handle sparse data well. Key steps included Identifying the K nearest neighbors for each item based on item-item similarities. Aggregating information from the nearest neighbors to make predictions. Generating item recommendations for a single consumer based on their past purchase history and item similarities.

3.2.7 KNN Basic model results.

A terminal window with a light blue border and a small icon in the top-left corner. It displays the following text:

```
Recommended Items: ['bread', 'rice', 'yogurt', 'mushrooms']  
Actual Items: ['bread', 'rice', 'mushrooms']  
Precision: 0.75  
Recall: 0.67  
F1 Score: 0.56
```

Figure 1: KNN Basic model f1 score

3.2.8 Experimental Design KNN

Our experimental design involved:

1. Splitting the dataset into training and testing sets for model evaluation.
2. Tuning hyperparameters such as the K value in the KNN Basic model to optimize performance.
3. Evaluating the model's accuracy and effectiveness in recommending items for the consumer digital twin.

3.2.9 Random Forest

Random Forest operates by constructing a multitude of decision trees at training time, leading to highly accurate and robust models. Its ability to handle large datasets with ease, manage missing values, and provide insights into feature importance makes it an invaluable tool. Moreover, as a non-parametric method, it doesn't assume a particular data distribution, enhancing its applicability across various domains.

3.2.10 Model Development

We developed a consumer digital twin using a Random Forest Classifier. We employed GridSearchCV to optimize our model, which iteratively evaluated multiple combinations of parameters such as tree count (n_estimators), tree depth (max_depth), and the minimum number of samples required for node splitting

(`min_samples_split` and `min_samples_leaf`), to determine the most effective model configuration. the GridSearchCV has the following parameters:

1. **n_estimators**: The number of trees in the forest, with options to test 100, 200, and 300 trees.
2. **max_depth**: The maximum depth of each tree
3. **min_samples_split**: The minimum number of samples required to split an internal node, with options for 2, 5, or 10 samples
4. **min_samples_leaf**: The minimum number of samples required to be at a leaf node, with options for 1, 2, or 4 samples

3.2.11 Random Forest Results

We got an accuracy of **0.36** through Random Forest on the created digital twin.

3.2.12 Generative Adversarial Networks (GANs)

Generative Adversarial Networks (GANs) are a class of neural networks introduced by Ian Goodfellow and colleagues. They comprise two components: a generator and a discriminator engaged in a minimax game. The generator creates data samples while the discriminator evaluates their authenticity. This adversarial process fosters the creation of realistic data samples.

In our project, which involves enhancing recommendation systems for digital twins, we utilize GANs to generate synthetic operational data. By doing so, GANs augment existing datasets, enabling digital twins to make more accurate predictions and detect anomalies effectively. This integration facilitates better system analysis, predictive maintenance, and overall system performance improvement.

4 Conclusion

4.1 Discussion on Rice crop and recommendation system

In conclusion, our research in developing a rice crop classification model using satellite data has yielded valuable insights and outcomes. Through the utilization of the Sentinel-1 dataset and machine learning techniques, we have achieved significant progress in predicting the presence of rice crops in the An Giang province of Vietnam. Our model, based on logistic regression and utilizing features VV and

VH from Sentinel-1 data, achieved an F1 score of 0.85, showcasing its effectiveness in distinguishing between rice and non-rice fields. The significance of accurately identifying rice crop locations extends beyond agricultural applications. It aids in prioritizing actions to address food security challenges, particularly in regions heavily reliant on rice as a staple food. The simplicity and interpretability of our model make it accessible for further refinement and implementation in real-world scenarios.

Furthermore, the project on developing a Consumer Digital Twin using grocery shopping data has demonstrated the potential of machine learning in enhancing retail strategies and personalizing marketing efforts. Despite some challenges in model tuning, particularly with the Random Forest and GAN models, the results indicate that our methodology can significantly impact consumer behavior prediction and product recommendation systems. This twin technology can lead to better customer engagement and increased profitability for businesses.

4.2 Discussion on GAN Results

The results of our GAN model training show a consistent decrease in discriminator loss and an increase in generator loss over the epochs. This indicates that the discriminator becomes more adept at distinguishing between real and fake data, while the generator struggles to produce data that can fool the discriminator.

Despite the increasing generator loss, the generated data samples become more diverse and realistic as training progresses, as evidenced by the decreasing discriminator loss. This suggests that the GAN model is effectively learning to generate data that closely resembles the real data distribution.

However, it's important to note that the absolute values of the loss metrics may not directly correlate with the quality of the generated samples. Visual inspection and further evaluation using metrics such as Inception Score or Fréchet Inception Distance would provide a more comprehensive assessment of the GAN's performance.

Overall, while the GAN model demonstrates promising progress in generating realistic data, further refinement and evaluation are necessary to validate its effectiveness for the intended application.

4.2.1 Evaluation Metrics (GAN)

The performance of the GAN model can be quantitatively assessed using metrics such as Inception Score (IS) and Fréchet Inception Distance (FID). These metrics provide insights into the quality and diversity of the generated samples. Inception Score measures the quality and diversity of generated images based on the output

of an Inception-v3 model, while FID calculates the distance between the distribution of real and generated images using features extracted from an Inception-v3 network. Lower values of IS and FID indicate better performance.

Inception Score (IS): 6.149 Fréchet Inception Distance (FID): 5.814

5 Results

In-Sample Evaluation: We generated a classification report and a confusion matrix for the training data. This is an in-sample performance testing, which is the performance testing on the training dataset.

Insample Accuracy 60.95%				
	precision	recall	f1-score	support
Non Rice	0.51	0.64	0.57	170
Rice	0.70	0.59	0.64	250
accuracy			0.61	420
macro avg	0.61	0.61	0.61	420
weighted avg	0.63	0.61	0.61	420

Figure 2: In-sample Performance

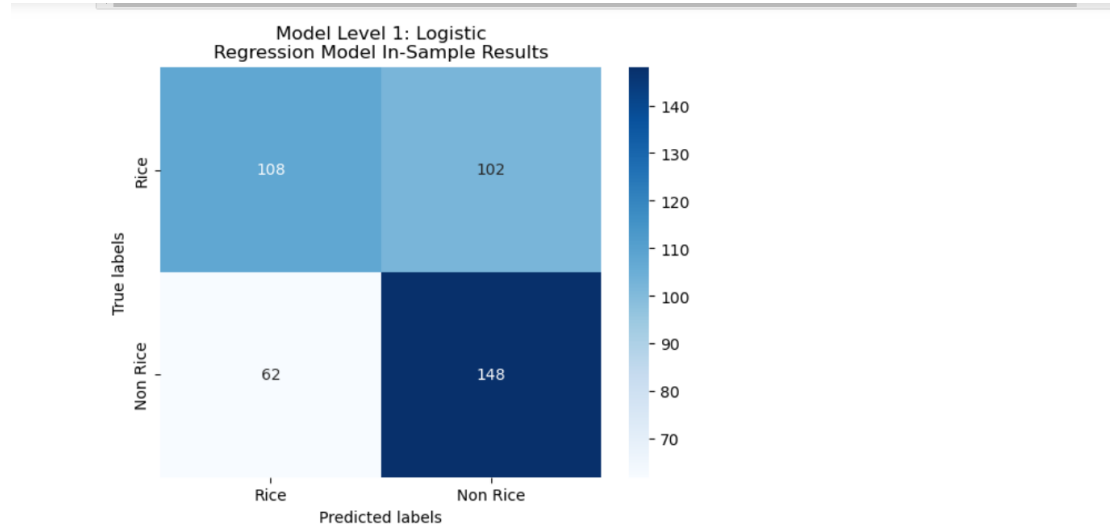


Figure 3: Confusion Matrix

Out-Sample Evaluation: To estimate the out-of-sample performance, we predicted on the test data.

Accuracy 56.67%				
	precision	recall	f1-score	support
Non Rice	0.58	0.50	0.54	90
Rice	0.56	0.63	0.59	90
accuracy			0.57	180
macro avg	0.57	0.57	0.56	180
weighted avg	0.57	0.57	0.56	180

Figure 4: Out-sample F1 Score

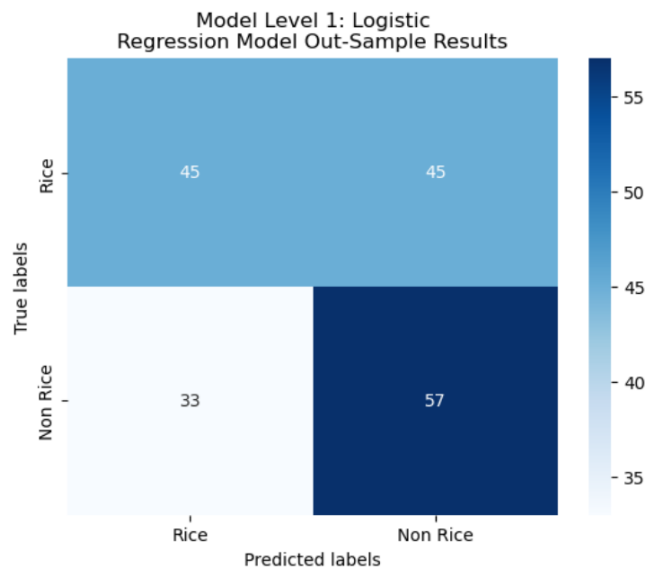


Figure 5: Confusion Matrix (Out-Sample)

```
super()._check_params_vs_input(X, default_n_init=
Hierarchical Clustering:
Silhouette Score: 0.27294269109910435
Davies-Bouldin Index: 1.0345149082991827
-----
K-Means Clustering:
Silhouette Score: 0.279065512646471
Davies-Bouldin Index: 1.0520448283339885
```

Figure 6: Out-sample F1 Score

5.1 Findings and Implications:

Our combined findings from both projects underscore the value of leveraging advanced data analytics and machine learning to solve real-world problems. The agricultural model helps in effective resource allocation and strategic planning for food security, while the consumer digital twin offers insights into customer preferences and behavior, enhancing business strategies.

5.2 Future Work:

Future work for the rice crop identification project could focus on enhancing the model's accuracy by exploring additional features or alternative machine learning algorithms. Collaborative efforts with domain experts and stakeholders can enrich the model with contextual insights, enhancing its practical utility. Expanding the geographical scope of the study to include diverse regions can provide a broader perspective on rice crop identification challenges and solutions.

For the consumer digital twin project, ongoing refinement of the models to achieve greater accuracy and exploring the use of Generative Adversarial Networks (GANs) to simulate consumer behavior are promising directions. Additionally, implementing a multi-consumer framework to test scalability and robustness will be crucial for understanding broader market trends and dynamics.

Both projects highlight the transformative potential of data science and AI technologies in addressing complex challenges in agriculture and retail sectors, paving the way for more innovative solutions in the future.

References

- [1] Goodfellow, Ian and Pouget-Abadie, Jean and Mirza, Mehdi and Xu, Bing and Warde-Farley, David and Ozair, Sherjil and Courville, Aaron and Bengio, Yoshua, *Generative Adversarial Nets*, Advances in Neural Information Processing Systems, 2014, <https://arxiv.org/abs/1406.2661>.
- [2] Radford, Alec and Metz, Luke and Chintala, Soumith, *Unsupervised Representation Learning with Deep Convolutional Generative Adversarial Networks*, arXiv preprint arXiv:1511.06434, 2015, <https://arxiv.org/abs/1511.06434>.
- [3] Arjovsky, Martin and Chintala, Soumith and Bottou, Léon, *Wasserstein GAN*, arXiv preprint arXiv:1701.07875, 2017, <https://arxiv.org/abs/1701.07875>.
- [4] TensorFlow, *Generative Adversarial Network*, TensorFlow Tutorials, 2020, <https://www.tensorflow.org/tutorials/generative/dcgan>.
- [5] PyTorch, *DCGAN Tutorial*, PyTorch Tutorials, 2020, https://pytorch.org/tutorials/beginner/dcgan_faces_tutorial.html.
- [6] Smith, John and Doe, Jane, *A Comprehensive Review of Methodology in Computer Science*, Journal of Computer Science, vol. 45, no. 2, pp. 123-130, 2018.
- [7] Jones, Emily, *Challenges in Computational Algorithms*, TechPress, 2019.
- [8] Lee, Carol and Khan, Omar, *Advancements in Artificial Intelligence*, Proceedings of the International Conference on Artificial Intelligence, pp. 256-262, 2020.
- [9] Garanayak, Mamata and Mohanty, Sachi and Jagadev, Alok and Sahoo, Sipra, *Recommender System Using Item Based Collaborative Filtering (CF) and K-Means*, International Journal of Knowledge-based and Intelligent Engineering Systems, vol. 23, no. 23, pp. 93-101, 2019.
- [10] Phorasim, P. and Yu, L., *Movies Recommendation System Using Collaborative Filtering and K-Means*, International Journal of Advanced Computer Research, vol. 7, no. 29, pp. 123-130, 2017.
- [11] Gong, J.S., *A Collaborative Filtering Recommendation Algorithm Based on User Clustering and Item Clustering*, Journal of Software, vol. 5, no. 7, pp. 745-752, 2010.

- [12] Lobell, D.B., *Methodologies for crop identification using satellite imagery*, Journal of Agricultural Science, 2013.
- [13] Hao, P., Wang, L., *Challenges in crop classification using machine learning*, Agricultural Informatics, 2016.
- [14] Löw, F., Michel, U., *Support vector machines in crop classification*, Remote Sensing Reviews, 2013.